20/21 Solutions

Sunday, November 7, 2021 8:49 PM

NANYANG TECHNOLOGICAL UNIVERSITY

SEMESTER I EXAMINATION 2020-2021

MH2500 – Probability and Introduction to Statistics

December 2020 TIME ALLOWED: 2 HOURS

INSTRUCTIONS TO CANDIDATES

1. This examination paper contains **SIX (6)** questions and comprises **FOUR (4)** printed pages.
2. Answer **ALL** questions. The marks for each question are indicated at the beginning of each question.
3. Answer each question beginning on a **FRESH** page of the answer book.
4. This is a **RESTRICTED OPEN BOOK** exam. Candidates are allowed **ONE** double-sided A4 size help sheet.
5. Candidates may use calculators. However, they should write down systematically the steps in the workings.
6. The table with the values of the standard normal distribution is included at the end of this examination paper.
7. All questions are the property of Nanyang Technological University.

QUESTION 1. **(20 marks)**

In a certain assembly plant, three machines B_1, B_2 and B_3 make 25%, 35% and 40%, respectively, of the products. It is known from past experience that 5%, 4% and 2% of the products made my each machine, respectively, are defective. Now, suppose a finished product is randomly selected.

(a) Find the probability that the selected product is a defective.

(b) Assume the selected product is a defective. Find the probability that this product was made by Machine B_1.

(c) If we know that the selected product is not produced by B_3, find the probability that it was produced by B_1.

(a) Let D be the event that product is a defective.

$$P(D) \overset{\text{LOTP}}{=} P(D|B_1)P(B_1) + P(D|B_2)P(B_2) + P(D|B_3)P(B_3)$$

$$= (0.05)(0.25) + (0.04)(0.35) + (0.02)(0.4)$$

$$= 0.0345 \,//$$

(b) $$P(B_1|D) = \frac{P(B_1 \cap D)}{P(D)} = \frac{P(D|B_1)P(B_1)}{P(D)}$$

$$= \frac{(0.05)(0.25)}{0.0345}$$

$$= \frac{25}{69} \,//$$

(c) $$P(B_1|B_3') = \frac{P(B_1 \cap \{B_3'\})}{P(B_3')}$$

Mutually exclusive

$$= \frac{P\{B_1 \cap (B_1 \cup B_2)\}}{P\{B_1 \cup B_2\}} = \frac{P(B_1)}{P(B_1)+P(B_2)} = \frac{0.25}{0.25+0.35}$$

$$= \frac{5}{12} \,//$$

QUESTION 2. **(15 marks)**

The serum cholesterol level X in 14-year-old boys has approximately a normal distribution with mean 170 and standard deviation 30.

(a) Find the probability that the serum cholesterol level of a randomly chosen 14-year-old exceeds 230.

(b) In a middle school there are 300 14-year-old boys. Find the probability that at least 8 boys have a serum cholesterol level that exceeds 230.

(a) $X \sim N(170, 30^2)$

$$P(X > 230) = P\left(\frac{X-170}{30} > \frac{230-170}{30}\right)$$
$$= P(Z > 2)$$
$$= 1 - \phi(2)$$
$$= 1 - 0.9772$$
$$= 0.0228$$

(b) Let Y be the r.v. denoting the # of boys that exceed 230 for its cholestrol level. $Y \sim Bin(300, 0.0228)$

$$P(Y \geq 8) = 1 - P(Y \leq 7)$$
$$= 1 - \sum_{i=0}^{7} \binom{300}{i} (0.0228)^i (1-0.0228)^{300-i}$$
$$= 0.3772$$

However, this approach is tedious in computation, $\therefore$ Normal Approximation can be used too. $Var(Y) = np(1-p) = 300(0.0228)(1-0.0228)$

$= 6.684048$, $E(X) = 6.84$

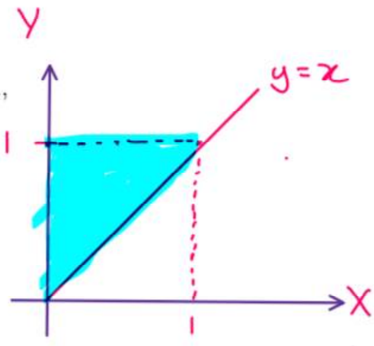
correction.

$$P(Y \geq 8) = P(Y > 7.5) = P\left(Z > \frac{7.5 - 6.84}{\sqrt{6.684048}}\right) = 1 - \phi(0.26)$$
$$= 0.3974$$

QUESTION 3. **(15 marks)**

The joint density of X and Y is given as:

$$f(x,y) = \begin{cases} \dfrac{1}{y}, \text{ if } 0 < x < y, \text{ and } 0 < y < 1, \\ \\ 0, \text{ elsewhere.} \end{cases}$$

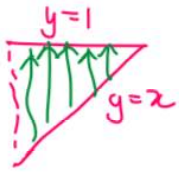


(a) Find the marginal distributions of X and Y, respectively.

(b) Are these two random variables X and Y independent?

(c) Find the probability $P\left\{X + Y > \dfrac{1}{2}\right\}$.

(a) $f_X(x) = \int_x^1 \frac{1}{y} \, dy$

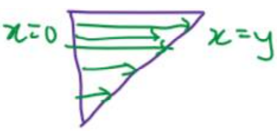


$= \left[\ln|y|\right]_x^1$

$= \ln 1 - \ln|x|$

$= -\ln x \quad , \quad 0 < x < 1.$

$f_Y(y) = \int_0^y \frac{1}{y} \, dx$

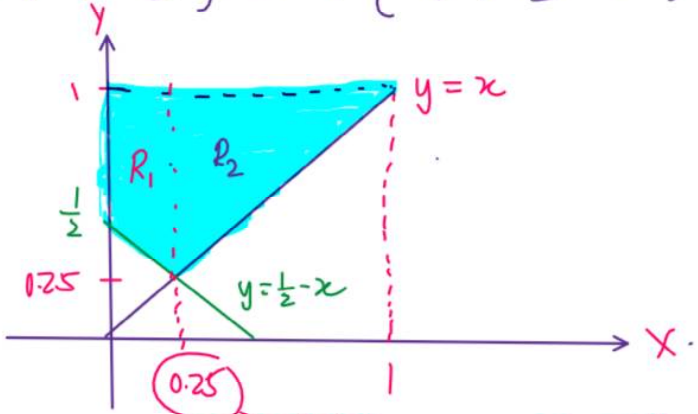


$= \frac{1}{y}\left[x\right]_0^y$

$= 1 \quad , \quad 0 < y < 1.$

(b) No. As $f_{XY}(x,y) = \frac{1}{y}$ but $f_X(x) f_Y(y) = -\ln x$,
Clearly $f_{XY} \neq f_X f_Y$. $\therefore$ X and Y are not independent.

(c) $P\{X+Y > \frac{1}{2}\} = P\{Y > \frac{1}{2} - X\}$



intersection of $y=\frac{1}{2}-x$ and $y=x$

Method 1) 'Split' into 2 regions.

$$P\{Y > \tfrac{1}{2} - X\} = \iint_{R_1} f(x,y)\,dy\,dx + \iint_{R_2} f(x,y)\,dy\,dx$$

$$= \int_0^{0.25}\int_{\frac{1}{2}-x}^{1} \frac{1}{y}\,dy\,dx + \int_{0.25}^{1}\int_{x}^{1} \frac{1}{y}\,dy\,dx$$

$$= \int_0^{0.25} \Big[\ln y\Big]_{\frac{1}{2}-x}^{1} dx + \int_{0.25}^{1} \Big[\ln y\Big]_{x}^{1} dx$$

$$= -\int_0^{0.25} \ln\left(\tfrac{1}{2}-x\right) dx - \int_{0.25}^{1} \ln x\,dx$$

$$\int_0^{0.25} \ln\left(\tfrac{1}{2}-x\right) dx = \int_0^{0.25} \ln\left(\frac{1-2x}{2}\right) dx$$

$$= \int_0^{0.25} \underbrace{1}_{v'}\,\underbrace{\ln(1-2x)}_{u}\,dx - \ln 2\,dx$$

$$= \Big[x\ln(1-2x)\Big]_0^{0.25} - \int_0^{0.25} \frac{-2}{1-2x}\,x\,dx - \Big[x\ln 2\Big]_0^{0.25}$$

$$= \frac{1}{4}\ln\frac{1}{2} + \int_0^{0.25} \frac{2x}{1-2x}\,dx - \frac{1}{4}\ln 2$$

$$= -\frac{1}{4}\ln 2 - \frac{1}{4}\ln 2 + \int_0^{0.25} -1 + \frac{1}{1-2x}\,dx$$

$$= \left[-x + \frac{\ln(1-2x)}{-2}\right]_0^{0.25} - \frac{1}{2}\ln 2 = -\frac{1}{4}$$

$$\int_{0.25}^{1} \ln x \, dx = \left[x \ln x\right]_{0.25}^{1} - \int_{0.25}^{1} \left(\frac{1}{x}\right) x \, dx$$

$$= -\frac{1}{4} \ln \frac{1}{4} - \left[x\right]_{0.25}^{1}$$

$$= -\frac{1}{4} \ln \frac{1}{4} - \frac{3}{4}$$

$$\therefore P\{Y > \tfrac{1}{2} - X\}$$

$$= \frac{1}{4} + \frac{3}{4} + \frac{1}{4} \ln \frac{1}{4}$$

$$= 1 + \frac{1}{2} \ln \frac{1}{2}$$

$$\approx 0.653$$

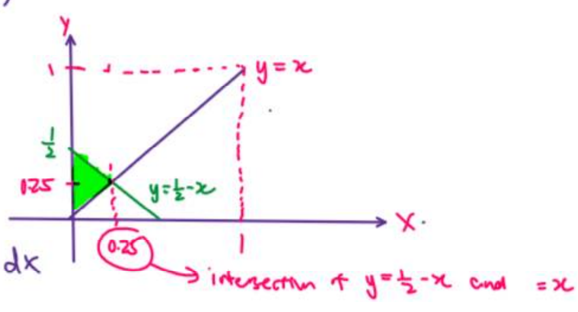
method2) 'complement' (only 1 region needed)

$$P\{Y > \tfrac{1}{2} - x\} = 1 - P\{Y \leq \tfrac{1}{2} - x\}$$

$$= 1 - \int_{0}^{0.25} \int_{x}^{\frac{1}{2} - x} \frac{1}{y} \, dy \, dx$$

$$= 1 - \int_{0}^{0.25} \left[\ln\left(\tfrac{1}{2} - x\right) - \ln x\right] dx$$

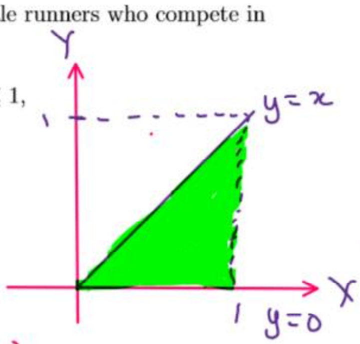
Same way as above



QUESTION 4. **(20 marks)**

The fraction X of male runners and the fraction Y of female runners who compete in marathon races are described by the joint density function

$$f(x,y) = \begin{cases} 8xy, & \text{if } 0 \le y \le x \le 1, \\ 0, & \text{elsewhere.} \end{cases}$$



(a) Find the covariance of X and Y.

(b) Find the correlation coefficient of X and Y.

(a) $\mathrm{Cov}(X,Y) = E(XY) - E(X)E(Y)$

$$E(XY) = \iint_{\mathbb{R}^2} xy\, f(x,y)\, dy\, dx$$
$$= \int_0^1 \int_0^x xy\,(8xy)\, dy\, dx$$
$$= 8\int_0^1 x^2 \int_0^x y^2\, dy\, dx$$
$$= 8\int_0^1 x^2 \left[\frac{y^3}{3}\right]_0^x dx$$
$$= \frac{8}{3}\int_0^1 x^5\, dx$$
$$= \frac{8}{3}\left[\frac{x^6}{6}\right]_0^1$$
$$= \frac{4}{9}$$

$$E(X) = \int_0^1 \int_0^x x\,(8xy)\, dy\, dx$$
$$= 8\int_0^1 x^2 \left[\frac{y^2}{2}\right]_0^x dx$$
$$= 4\int_0^1 x^4\, dx$$
$$= 4\left[\frac{x^5}{5}\right]_0^1$$
$$= \frac{4}{5}$$

$$E(Y) = \int_0^1 \int_0^x y\,(8xy)\, dy\, dx$$
$$= 8\int_0^1 x \left[\frac{y^3}{3}\right]_0^x dx$$
$$= \frac{8}{3}\int_0^1 x^4\, dx$$
$$= \frac{8}{3}\left[\frac{x^5}{5}\right]_0^1$$
$$= \frac{8}{15}$$

$$\therefore \mathrm{Cov}(X,Y) = \frac{4}{9} - \frac{4}{5} \times \frac{8}{15} = \frac{4}{225}$$

(b) Method 1: Integrate with joint pdf

$$E(X^2) = \int_0^1 \int_0^x x^2 f(x,y)\,dy\,dx$$
$$= \int_0^1 \int_0^x x^2 (8xy)\,dy\,dx = \frac{2}{3}$$

$$E(Y^2) = \int_0^1 \int_0^x y^2 f(x,y)\,dy\,dx$$
$$= \int_0^1 \int_0^x y^2 (8xy)\,dy\,dx = \frac{1}{3}$$

Method 2: Find marginals of X and Y.

$$f_X(x) = \int_0^x 8xy\,dy = 8x\left[\frac{y^2}{2}\right]_0^x = 4x^3, \text{ if } 0<x<1$$
$$f_Y(y) = \int_y^1 8xy\,dx = 8y\left[\frac{x^2}{2}\right]_y^1 = 4y(1-y^2)$$
if $0<y<1$.

Then find $E(X)$, $E(Y)$, $E(X^2)$, $E(Y^2)$ using

$$\int_0^1 x f_X(x)\,dx \quad \int_0^1 y f_Y(y)\,dy \quad \int_0^1 x^2 f_X(x)\,dx \quad \int_0^1 y^2 f_Y(y)\,dy.$$

$$Var(X) = E(X^2) - \{E(X)\}^2 = \frac{2}{75}$$
$$Var(Y) = E(Y^2) - \{E(Y)\}^2 = \frac{11}{225}$$

$$\rho = \frac{Cov(X,Y)}{\sqrt{Var(X)\,Var(Y)}}$$

$$\therefore \rho = \frac{\frac{4}{225}}{\sqrt{\frac{2}{75}}\sqrt{\frac{11}{225}}} \approx 0.492$$

QUESTION 5. **(15 marks)**

An urn has n white and m black balls that are removed one at a time in a randomly chosen order. Find the expected number of instances in which a white ball is immediately followed by a black one.

Let X_i be the indicator variable that the ith draw is a Black ball and $(i-1)^{th}$ draw is a White ball. $(2 \le i \le n+m)$

Let $X = X_2 + X_3 + \cdots + X_{n+m}$ (Note: this instance cannot occur at the 1st trial).

$$E[X_i] = P\{ i^{th} \text{ draw is black, } (i-1)^{th} \text{ draw is white} \}$$

$$= \frac{m}{n+m} \times \frac{n}{n+m-1}$$

$$= \frac{nm}{(n+m)(n+m-1)}$$

By linearity, $$E(X) = \sum_{i=2}^{n+m} E[X_i]$$

$$= (n+m-2+1) \times \frac{nm}{(n+m)(n-m+1)}$$

$$= (\cancel{n+m-1}) \times \frac{mn}{(n+m)(\cancel{n-m+1})}$$

$$= \frac{mn}{n+m}$$ //

QUESTION 6. **(15 marks)**

An instructor has 50 exams that will be graded in sequence. The times required to grade the 50 exams are independent, with a common distribution that has mean 20 minutes and standard deviation 4 minutes. By applying the Central Limit Theorem, approximate the probability that the instructor will grade at least 25 of the exams in the first 450 minutes of work.

Let X_i be the time taken to grade the i^{th} exam.

$E[X_i] = 20, \quad Var[X_i] = 4^2$

Let $T = \sum_{i=1}^{25} X_i$

$E(T) = \sum_{i=1}^{25} E[X_i] = \sum_{i=1}^{25} 20 = 500$

$Var(T) = \sum_{i=1}^{25} Var(X_i) = 25 \times 4^2 = 400$

As X_i's are independent.

$$P(T \leq 450) \overset{CLT}{=} P\left(\frac{T - E(T)}{\sqrt{Var(T)}} \leq \frac{450 - 500}{\sqrt{400}}\right)$$

$$= P\left(Z \leq \frac{-50}{20}\right)$$

$$= P(Z \leq -2.5)$$

$$= 1 - \phi(2.5)$$

$$= 1 - 0.9938$$

$$\approx 0.0062$$

END OF PAPER